

SMART INDIA HACKATHON 2026

- **Problem Statement ID** – PS 26105
- **Problem Statement Title** – AI-Powered Continuous Cyber Risk Quantification and Investment Optimization Platform

Theme – Blockchain & Cybersecurity

- **PS Category** – Software
- **Team ID** –
- **Team Name** – Private Keys



CyberRisk: AI-Powered Cyber Risk Quantification & Optimization Platform

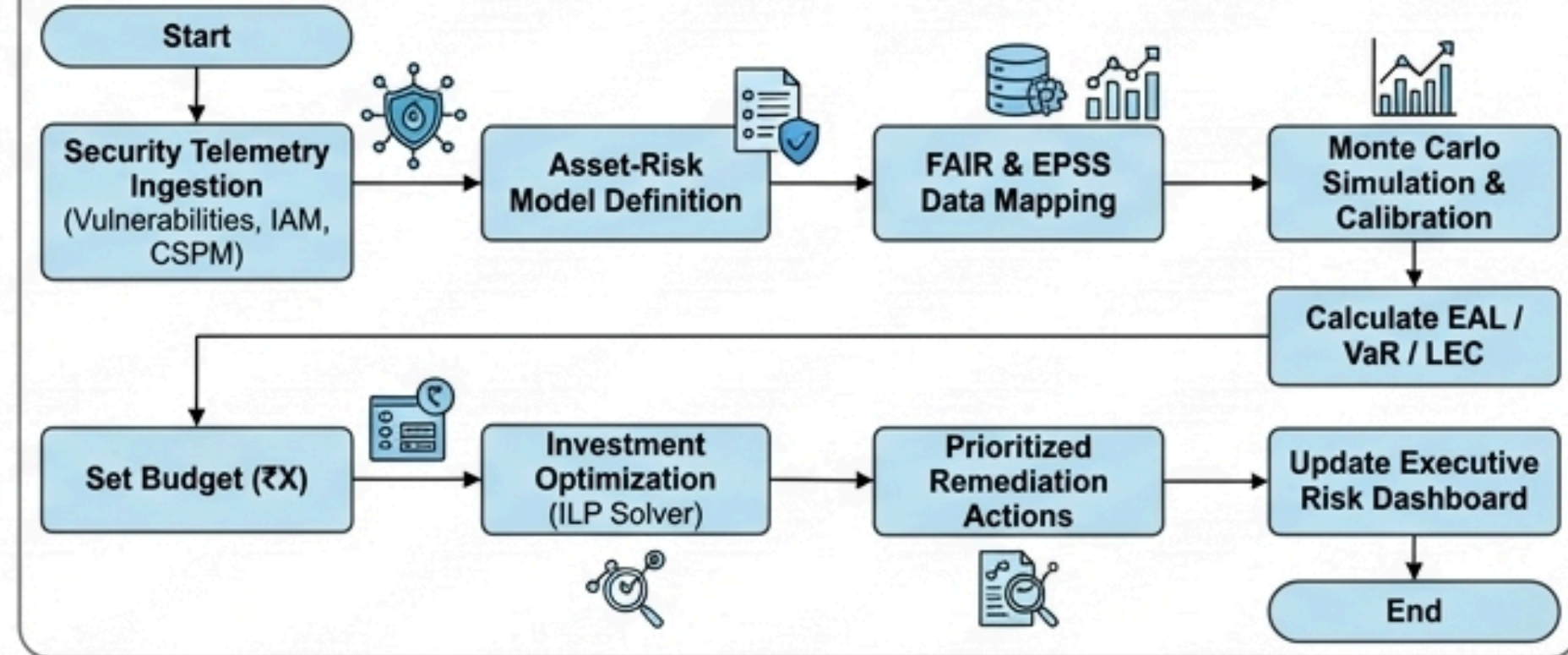
Converting Technical Findings into Budgetable Financial Exposure



Proposed Solution / Approach

- **Security Telemetry:** Integrates **Vulnerabilities · IAM · SIEM · EDR · CSPM** into a unified asset-risk model.
- **Risk Quantification:** **FAIR + EPSS + CVSS + Controls** to convert technical findings into monetary cyber risk.
- **Risk Intelligence:** **Monte Carlo simulation** to calculate **EAL · VaR · LEC** at asset, business-unit and enterprise levels.
- **Investment Optimizer:** **ILP-based optimization** to select security controls delivering maximum modeled risk reduction under **₹X budget**.
- **Control Impact:** Compares **EAL Before vs EAL After** to quantify control effectiveness, risk reduction and ROSI.
- **AI Risk Copilot:** Plain-English queries over computed risk data for faster executive and technical decision-making.
- **What-If Simulation:** Models the financial impact of remediation, MFA, security controls, and delayed actions.

CyberRisk Platform End-to-end Workflow



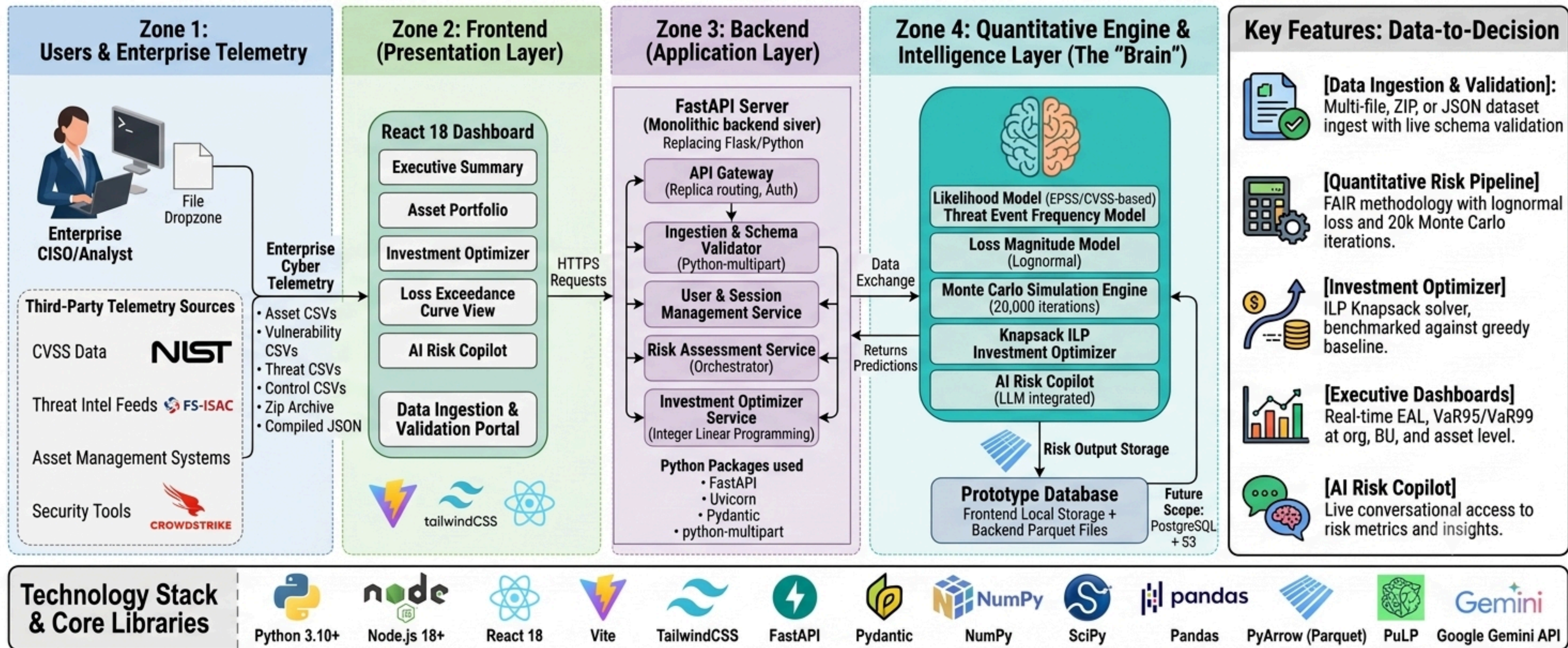
Innovation and Uniqueness



TECHNICAL APPROACH: CyberRisk - FAIR-Based Platform

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FEASIBILITY AND VIABILITY

Feasibility

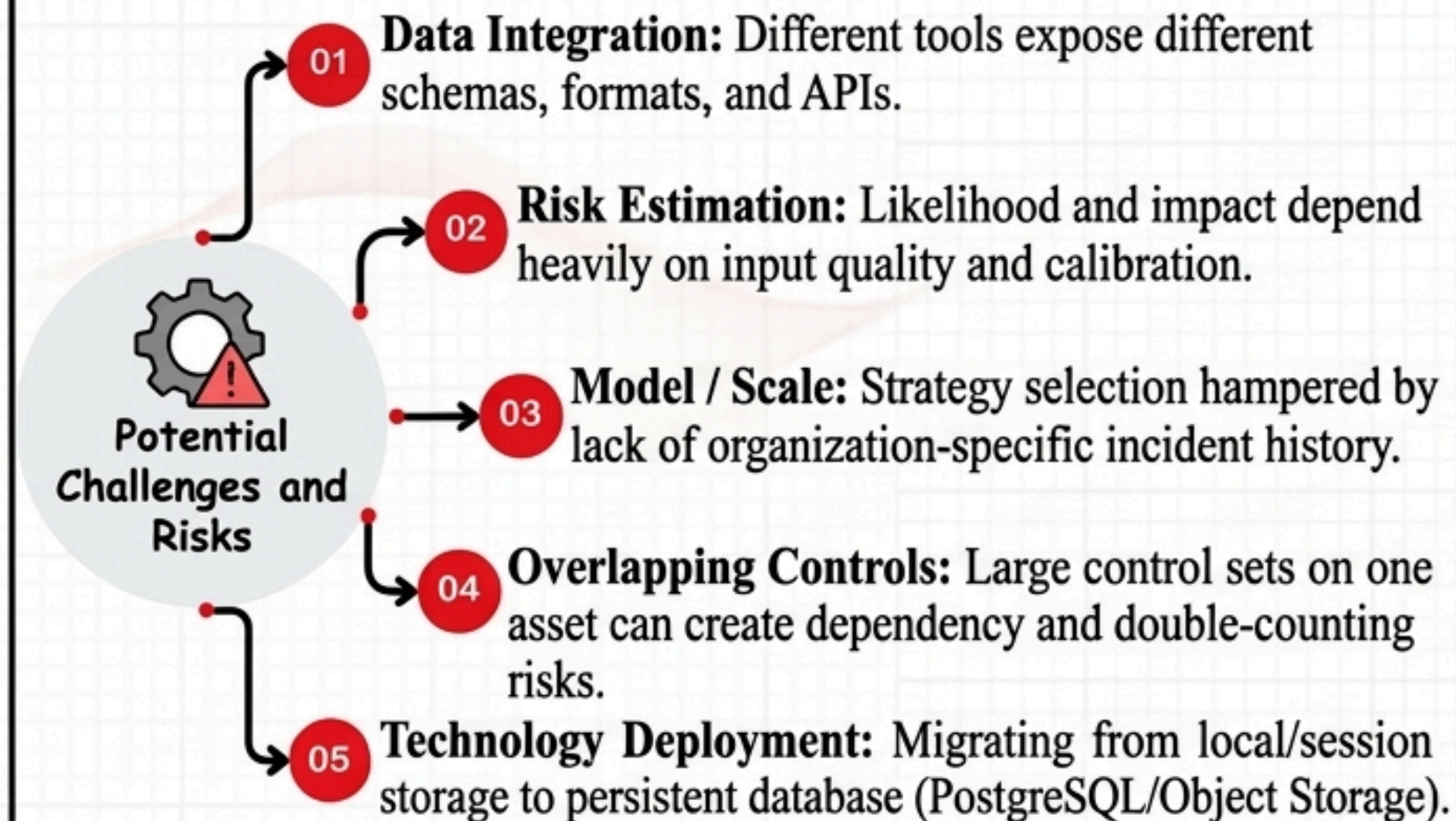
- Proven Open FAIR™ framework & Monte Carlo simulations for quantitative financial risk assessment.
- Modular architecture using Python, FastAPI, and a stable schema, tested on synthetic CVE/CVSS data.
- Standard PostgreSQL support for modular and scalable deployment.

Viability

- Quantifies actual monetary value (e.g., EAL, $\text{VaR}_{5/99}^9$) instead of vague 'High/Medium/Low' labels.
- ROI (Return on Security Investment) computation identifies cost-effective controls for senior management and boards.
- Addresses misprioritized remediation by calculating mathematically optimal security investment plans.

Practical Implementation

- Complete source code provided for Python backend (risk engine, API) and React frontend (dashboard, chatbot).
- Tested ILP optimizer identifies optimal control sets under budget constraints, outperforming greedy baselines.
- Interactive dashboard with executive and technical views and a natural-language AI Risk Copilot.



Normalized Schema: Map all telemetry sources into one unified asset-risk model before calculation.

Explainable Quantification: Retain FAIR inputs and intermediate values; expose EAL-EAL-after ROSI re-simulation.

Simulation + Testing: Use synthetic/demo inputs during development; recalibrate with organizational data.

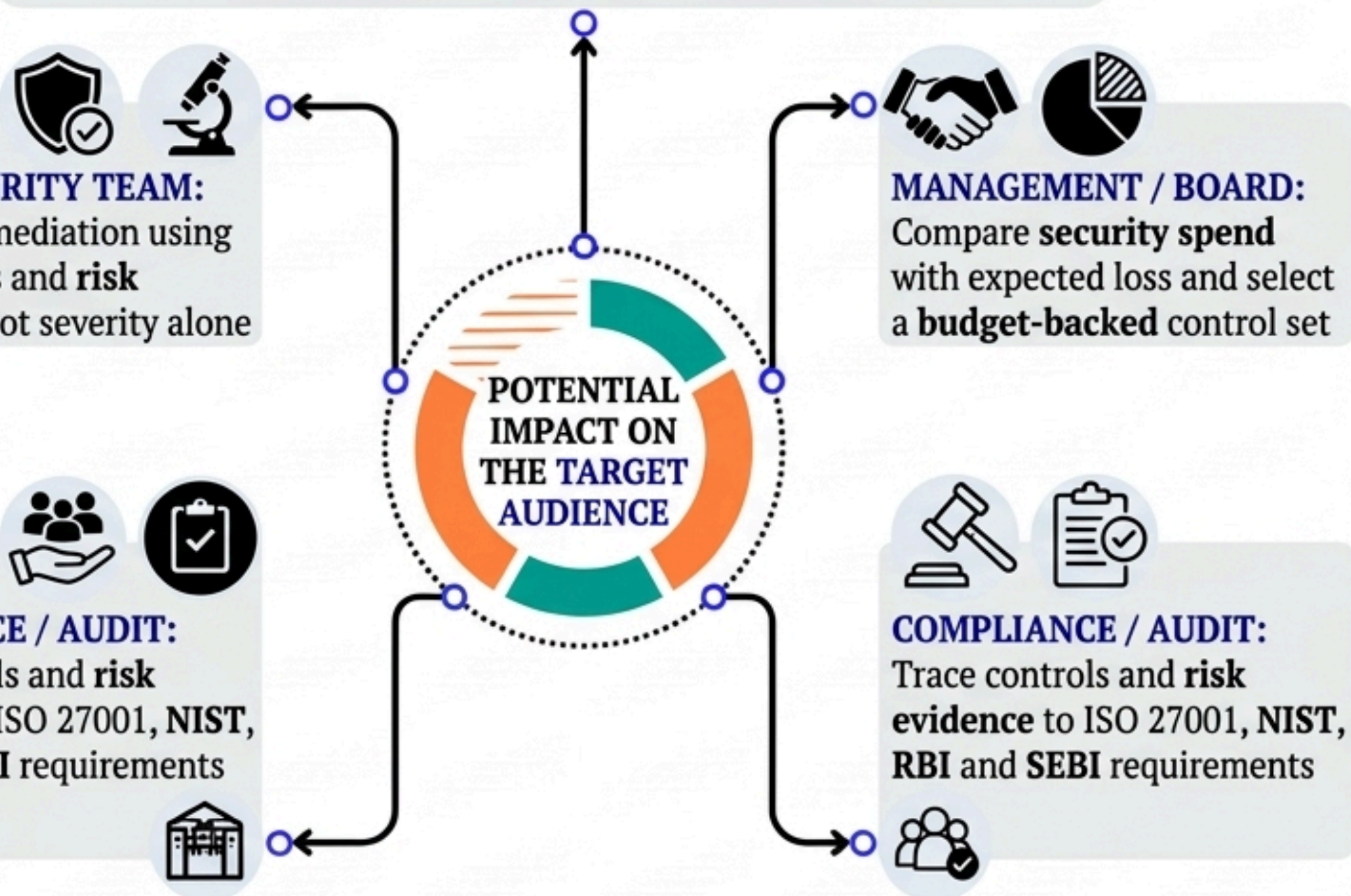
Represent Dependencies: Explicitly represent and block incompatible or duplicate control actions in the ILP solver.

Flexible Backend: Support hybrid PostgreSQL + analytical file storage (e.g., Parquet, Object Storage).



IMPACT AND BENEFITS

CyberRisk - FAIR-Based Cyber Risk Quantification & Investment Optimization Platform



Benefits of the solution

“Provides a **single view** of risk, controls, and **financial exposure**; traces to framework clauses; and generates **exportable** risk/compliance reports.”



“Identifies high-contributing assets and controls; preserves **EAL-before / EAL-after** for actions; and prioritizes remediation from **enterprise** to **asset level**.”

“Transforms **expected loss** into a budgetable quantity; **compares** control **cost** with modeled reduction; and enables “**best controls under ₹X**” scenarios.”



“India’s first quantitative risk management platform—real visibility, real control, real resilience.”

Methodology & Standards Foundations

- **FAIR Institute – Factor Analysis of Information Risk (Open FAIR):** Core quantitative risk taxonomy underlying the platform's Monte Carlo risk-quantification engine. <https://www.fairinstitute.org/what-is-fair>
- **The Open Group – Open FAIR Body of Knowledge (O-RA / O-RT Standards):** Formal risk analysis and risk taxonomy standards referenced for the likelihood, frequency, and loss-magnitude modeling. <https://www.opengroup.org/open-fai>
- **FIRST.org – Exploit Prediction Scoring System (EPSS):** Daily-updated exploitation-probability scores used as the likelihood input for the vulnerability layer. <https://www.first.org/epss/>

Regulatory & Compliance References (India)

- **Reserve Bank of India – Cyber Security Framework in Banks (2016):** Regulatory basis for board-level, financially-quantified cyber risk reporting in Indian BFSI institutions. <https://www.rbi.org.in/commonman/english/scripts/Notification.aspx?Id=1721>
- **SEBI – Cybersecurity and Cyber Resilience Framework (CSCRF), Aug 2024:** Governance-Identify-Protect-Detect-Respond-Recover structure informing the platform's compliance-mapping roadmap. [https://nsdl.co.in/downloadables/pdf/2024-0118-Policy-SEBI Circular on Cybersecurity and Cyber Resilience Framework \(CSCRF\) for SEBI Regulated Entities \(REs\).pdf](https://nsdl.co.in/downloadables/pdf/2024-0118-Policy-SEBI_Circular_on_Cybersecurity_and_Cyber_Resilience_Framework_(CSCRF)_for_SEBI_Regulated_Entities_(REs).pdf)

Engineering & Quantitative Tooling

- **NIST Cybersecurity Framework (CSF) 2.0:** Six-function (Govern-Identify-Protect-Detect-Respond-Recover) reference model used for control categorization. <https://www.nist.gov/cyberframework>
- **PuLP / COIN-OR CBC Solver Documentation:** Open-source Integer Linear Programming library powering the Knapsack ILP investment optimizer. <https://coin-or.github.io/pulp/>